

HAT-P-19b

R-1134

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241028 · created 2026-07-30T23:22:49+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460611.6323 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.014
Transit depth (Rp/Rs)^2	3.25 +/- 0.52 [%]
Orbital Inclination (inc)	89.71 +/- 0.7
Airmass coefficient 1 (a1)	1.067 +/- 0.015
Airmass coefficient 2 (a2)	-0.06 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	None
Optimal Aperture	3.69
Optimal Annulus	15.25
Transit Duration (day)	0.0987 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.014 vs 0.1418 (deviation 2.73σ)
Duration fit vs published	0.0987 d vs 0.1196 d (deviation 5.5σ)
Mid-time O–C	3.8 min
Depth SNR	12.9
Residual scatter	1.38 %

Light curve

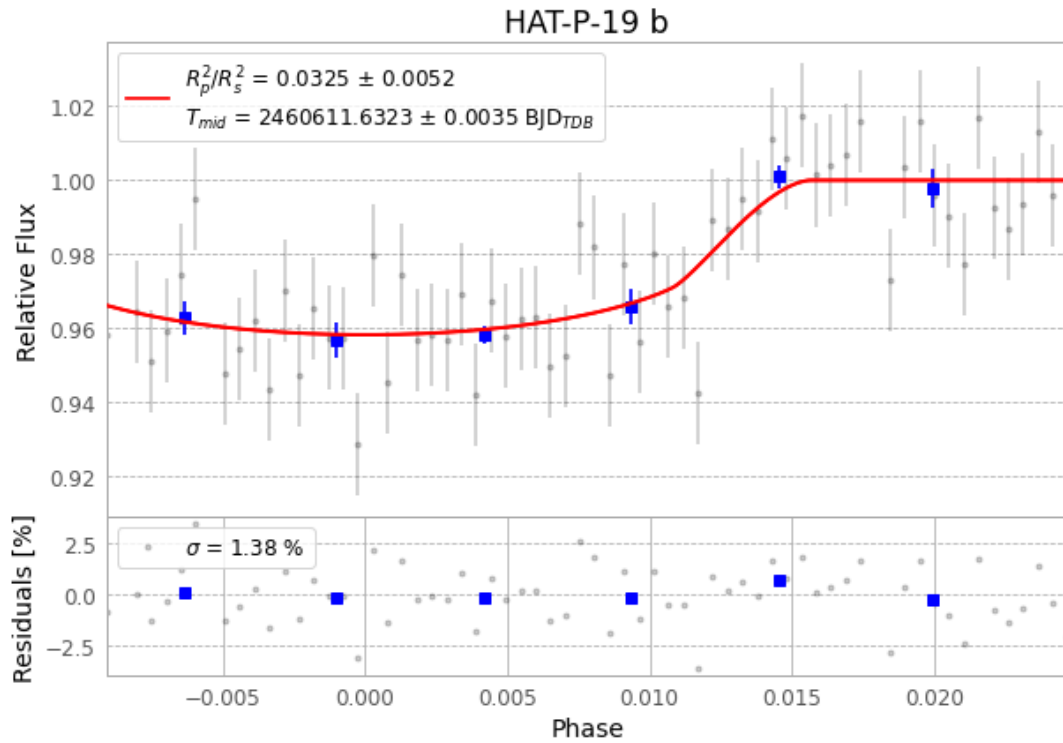


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-10-27.png (SHA-256 878117a604c65e7c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 188], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a4fa35f92cd9eeb4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), HATP-19241028020917.FITS → HATP-19241028052415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241028.log (b780d5432a3f...), FinalLightCurve_HAT-P-19 b_2024-10-27.png (878117a604c6...), FinalLightCurve_HAT-P-19 b_2024-10-27.csv (75dd1ab266f1...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:22Z.